

Data Cleaning & Reporting Automation

Data Analytics Internship Project

Objective

To automate data cleaning and reporting processes using Python and data analysis tools.

Abstract

Data quality is critical for accurate analysis. This project demonstrates automated techniques for handling missing values, removing duplicate records, and generating summary reports.

Tools Used

Python, Pandas, NumPy

Methodology

1. Load Dataset
2. Identify Missing Values
3. Remove Duplicate Records
4. Standardize Data
5. Generate Automated Reports
6. Export Clean Dataset

Sample Dataset

The sample dataset contains customer information with missing values and duplicate records for cleaning and analysis.

Results

The automation process improved data quality by eliminating inconsistencies and generating summary statistics automatically.

Conclusion

Automated data cleaning reduces manual effort, improves efficiency, and ensures reliable reporting for business decisions.